

Conclusion

Lean is an **efficient programming language** and **proof assistant**.

Lean is very extensible and is implemented in Lean.

Lean proofs are maintainable, stable, and transparent.

Progress is accelerating with the Lean FRO: module system, new compiler, new proof automation, etc.

The Mathlib community is changing how math is done.

AI + Lean will have a huge impact on mathematics and software/hardware verification.

It is not just about proving but also understanding complex objects and proofs, getting new insights, and navigating through the “thick jungles” that are **beyond our cognitive abilities**.

